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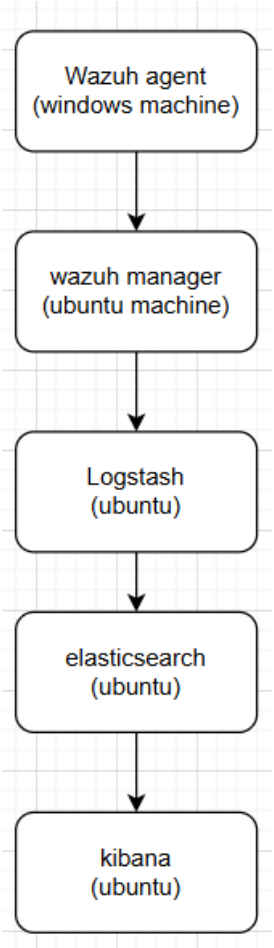
25/02/26

SOC Task 1 Report

I will use ubuntu virtual machine and windows host for a SOC lab. ELK (elastic, logstash, kibana) stack is already installed in ubuntu. Now I will install wazuh agent in windows and wazuh manager in ubuntu. The pipeline will be



First, we collect information from windows, and with the help of wazuh manager in ubuntu, we transfer that information to ubuntu. Now those information logs are processed by logstash and will be sent to elasticsearch then in kibana dashboard we can see those logs.

Install wazuh agent in windows then after installation type the manager IP which is ubuntu's IP address in this case then to get authentication key, type the command in ubuntu which is

- sudo /var/ossec/bin/manage_agents

type E, then provide the id of the agent. Now in windows, run powershell as admin, restart wazuh service and make sure that he service is running.

Now, write a code in the path /etc/logstash/conf.d
Create a .conf file by the command

- nano <name>.conf

in that conf file, I entered a code which is given below.

Code(wazuh.conf)

```
input {
  # If reading from the Wazuh alerts file
  file {
    path => "/var/ossec/logs/alerts/alerts.json"
    codec => "json"
    start_position => "beginning"
  }
}

filter {
  # Normalize the timestamp to Elastic format
  date {
    match => [ "timestamp", "ISO8601" ]
    target => "@timestamp"
  }

  # Flatten nested Windows data for easier searching in Kibana
  if [data][win][system] {
    mutate {
      add_field => { "event_id" => "%{[data][win][system][eventID]}" }
      add_field => { "channel" => "%{[data][win][system][channel]}" }
    }
  }
}

output {
  elasticsearch {
    hosts => ["http://localhost:9200"]
    index => "wazuh-normalized-%{+YYYY.MM.dd}"
    data_stream => false
    user => "elastic"
    password => "-xp4WxZsEEQeb1c0C+Ll"
  }
}
```

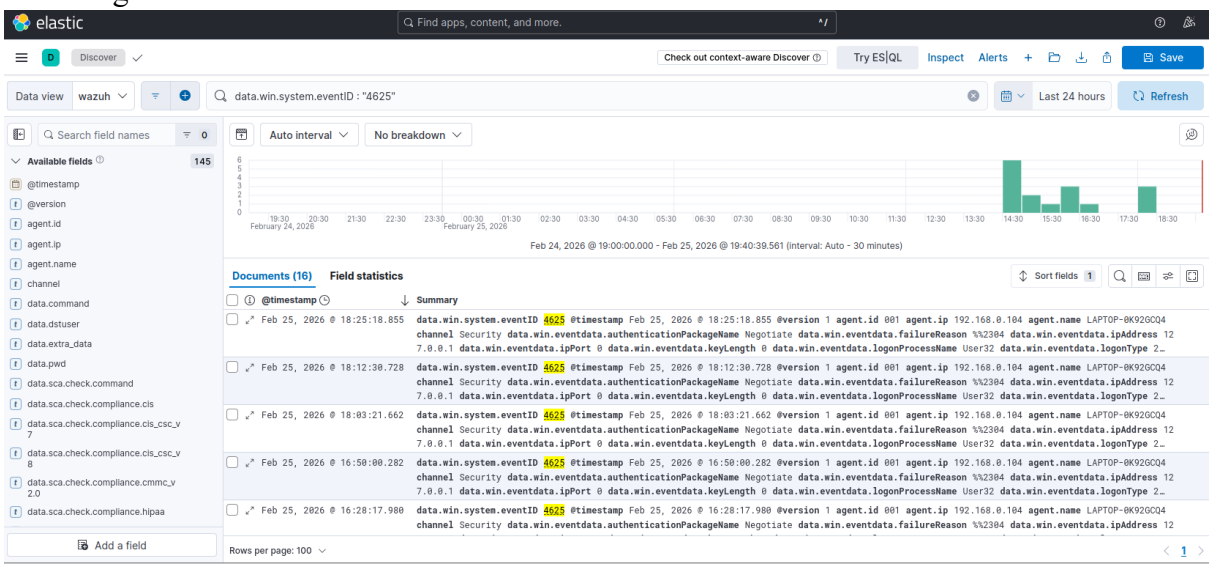
Save it and disable, then enable logstash and then start the logstash again. /var/ossec/logs/alerts/alerts.json, give permissions so that logstash can read this file, for that I entered the command

- sudo chmod 644 /var/ossec/logs/alerts/alerts.json

now in kibana dashboard, go to stack management, then go to data views, then click on create data view, fill the details then click on save data view to kibana. Now go to discover and you can see the logs. You can filter those logs using KQL query. I typed the command

- data.win.system.eventID : "4625"

in kql query box in kibana. The event id 4625 refers to wrong password entered which is failed logins.



For apache logs create another conf file in /etc/logstash/conf.d
Code for gathering apache logs is

Code(apache.conf)

```
input {
  file {
    path => "/var/log/apache2/access.log"
    start_position => "beginning"
    sincedb_path => "/dev/null" # For testing: re-reads the file from the start
  }
}

filter {
  # Parse the standard Apache 'Combined' or 'Common' log format
  grok {
    match => { "message" => "%{COMBINEDAPACHELOG}" }
```

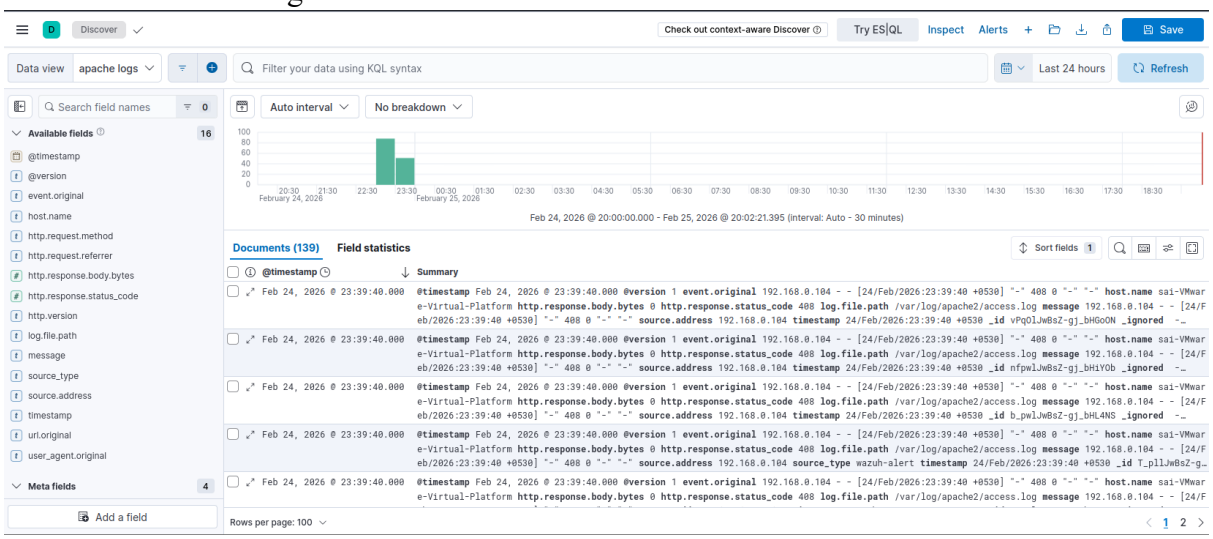
```
  }

  # Convert the string-based timestamp into a proper Logstash date object
  date {
    match => [ "timestamp", "dd/MMM/yyyy:HH:mm:ss Z" ]
    target => "@timestamp"
  }
}

output {
  # Print the normalized JSON to the terminal for verification
  stdout {
    codec => json_lines
  }

  # Send the normalized JSON to Elasticsearch
  elasticsearch {
    hosts => ["http://localhost:9200"]
    index => "apache-logs-%{+YYYY.MM.dd}"
  }
}
```

Now if there are 2 or more conf files, I typed a command so that there will be only one conf file which is

- sudo mv /etc/logstash/conf.d/apache.conf /etc/logstash/conf.d/apache.conf.bak

I did this because pipelines are mixing, like im getting apache logs in wazuh. And before typing this command always stop and disable logstash. after typing this command enable and start the logstash.

I got some apache logs where windows connecting to apache server in ubuntu, below is the screenshot of those logs in kibana.



Practical tasks

To install snort in ubuntu, type the following command which is,

- sudo apt-get install snort

During installationit asks for the ip address, you can type your ip address with CIDR number, For example, in my case it is 192.168.0.0/24

After installing snort go to path /etc/snort
then type the command

- sudo nano snort.conf

in that .conf file change ipvar HOME_NET <ipaddress>
in <ip address> type your ip address.
Next comment all rules except local.rules because in local.rules you can enter your own rules.
Now, write a rule to detect HTTP requests to "malicious.com"
type the rule in /etc/snort/rules/local.rules and the rule is given below

- alert tcp any any <> any 80 (msg:"Malicious Domain"; content:"malicious.com"; http_header; sid:1000001;)

save local.rules and type the command

- sudo snort -T -c /etc/snort/snort.conf -A console -i ens33

if the output does not contain any error type the command

- sudo snort -c /etc/snort/snort.conf -A console -i ens33

Now open another terminal while that command is running and enter the below command in that another terminal which is,

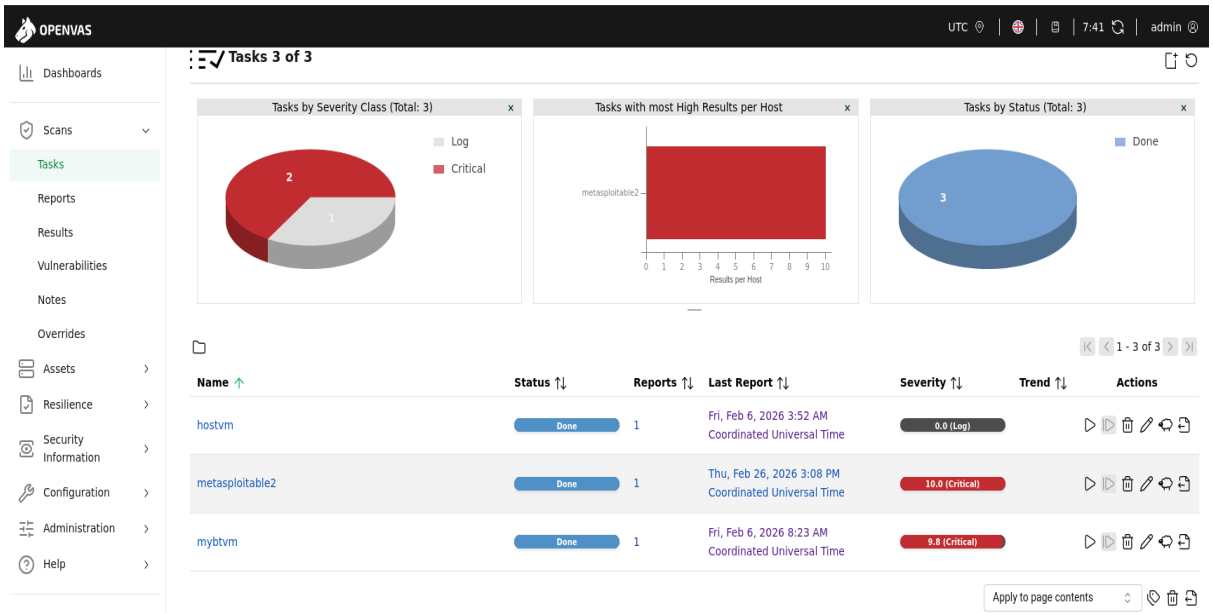
- curl http://malicious.com

then in this terminal you will get an output like below,

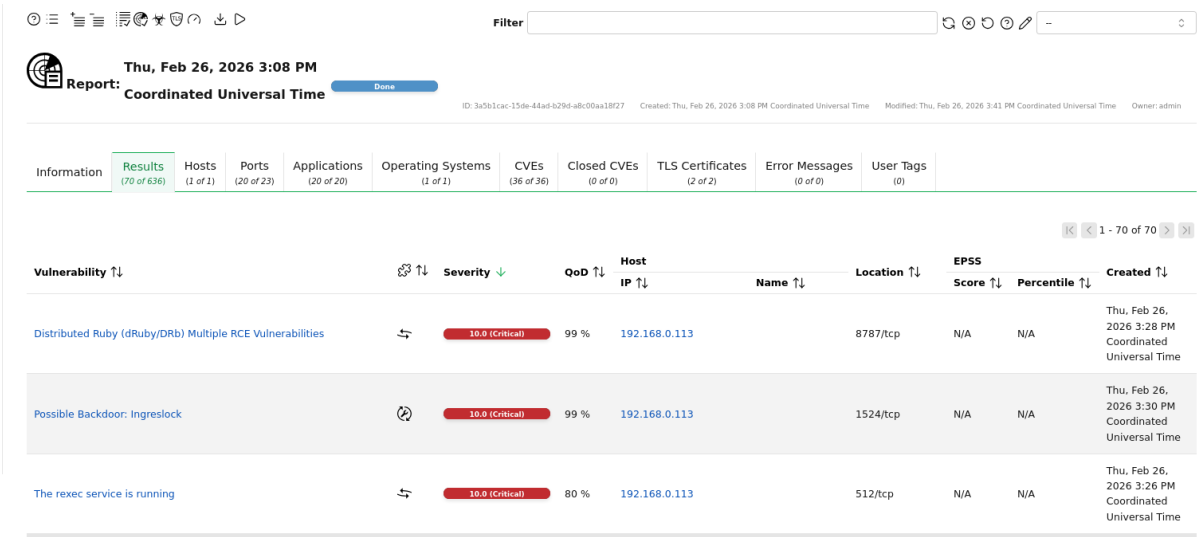
```
sai@sai-VMware-Virtual-Platform:/etc/snort$ sudo snort -c /etc/snort/snort.conf -A console -i ens33 -q
02/26-17:00:53.576480  [**] [1:1000001:0] Malicious Domain [**] [Priority: 0] {TCP} 149.28.227.54:80 -> 192.168.0.111:35588
```

Using openvas to detect vulnerabilities in vulnerable virtual machine

I used openvas instead of nessus because openvas is free and I already have it. Now I installed metasploitable2 in my virtual box to scan the vulnerabilities in it by using openvas and nmap.
After using nmap to get the ip address of the metasploitable2, I started openvas and scanned Metasploitable2 and after some time the scan got finished. The output screenshot of it is,



Metasploitable2 machine has severity level 10 is vulnerable. And when I clicked on the report, the output is



I downloaded this report a PDF and the first top 3 vulnerabilities are,

- Distributed Ruby (dRuby/DRb) Multiple RCE Vulnerabilities
- Possible Backdoor: Ingreslock
- The rexec service is running

These 3 vulnerabilities have CVSS score of 10 which is high.

Osquery monitoring

I installed windows 10 virtual machine which runs on virtualbox. After opening windows, I first installed chocolatey package, and for that I typed the command in windows powershell as admin, and the command is

- Set-ExecutionPolicy Bypass -Scope Process -Force; [System.Net.ServicePointManager]::SecurityProtocol = [System.Net.ServicePointManager]::SecurityProtocol -bor 3072; iex ((New-Object System.Net.WebClient).DownloadString('https://community.chocolatey.org/install.ps1'))

Now after this, I installed osquery by typing the command

- Choco install osquery

Type y for full installation.

After installing osquery go to the path where osquery is installed in the powershell, for me it is
C:\Program Files\osquery
After that type the command below to execute osquery

- ./osqueryi.exe

```
PS C:\Program Files\osquery> .\osqueryi.exe
Using a ▯[1mvirtual database▯[0m. Need help, type '.help'
osquery>
```

Now minimize the powershell and create a batch file, to do this open notepad and type
@echo off
echo Malicious process simulation running...
timeout /t 15
after typing the above 3 lines, save the file and in save as type select All Files. The name should end with .bat.
My batch file name is fake_malware.bat
Now by clicking on that batch file you will get something like this

```
C:\Windows\system32\cmd.exe
malicious process simulation running

Waiting for  7 seconds, press a key to continue ...
```

While it is running, go the powershell in which you opened osquery and type the command

- SELECT pid, name, cmdline, path FROM processes WHERE cmdline LIKE '%fake_malware%';

In place of fake_malware use your batch file name.

That command output is

```
osquery> SELECT pid, name, cmdline, path FROM processes WHERE cmdline LIKE '%fake_malware%';
+------+---------+-----------------------------------------------------------------------------------+----------------------------+
| pid  | name    | cmdline                                                                           | path                       |
+------+---------+-----------------------------------------------------------------------------------+----------------------------+
| 2464 | cmd.exe | C:\Windows\system32\cmd.exe /c ""C:\Users\vboxuser\Desktop\fake_malware.bat" " | C:\Windows\System32\cmd.exe |
+------+---------+-----------------------------------------------------------------------------------+----------------------------+
osquery>
```

And if you type the command below then the output will be

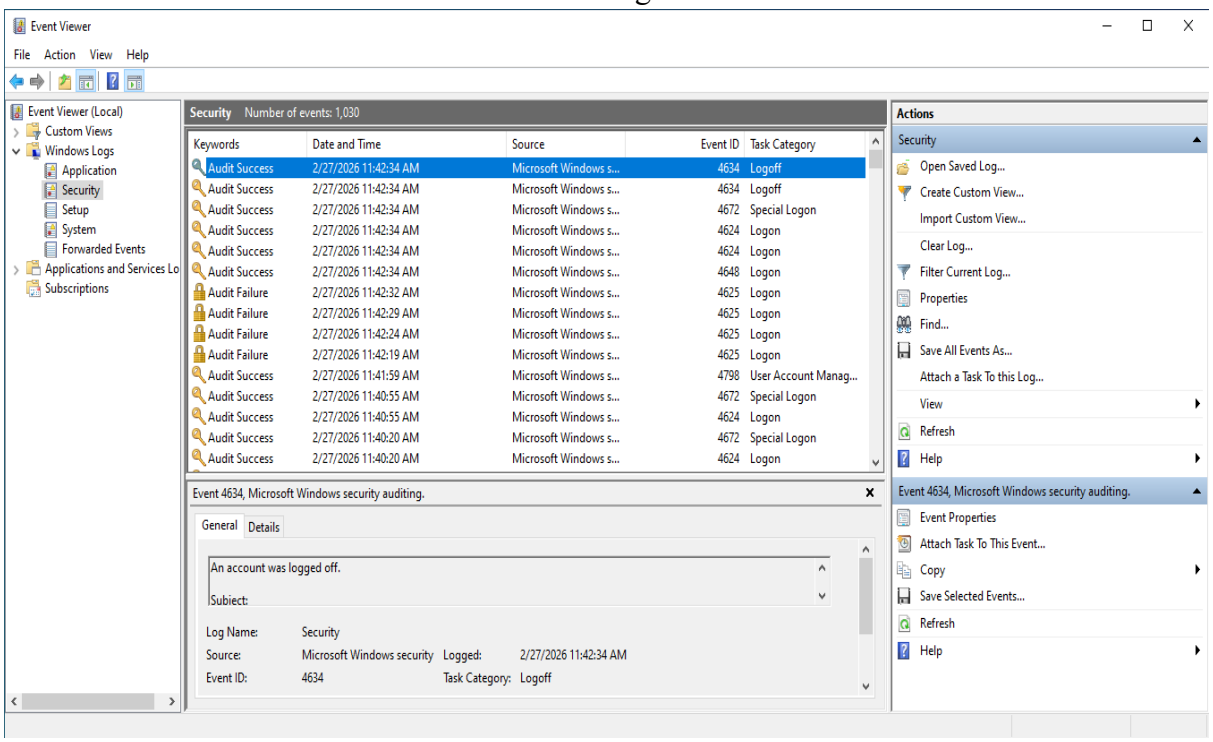
- SELECT * FROM processes WHERE cmdline LIKE '%fake_malware%';

Above command output

```
osquery> SELECT * FROM processes WHERE cmdline LIKE '%fake_malware%';
+------+---------+----------------------------+-----------------------------------------------------------------------------------+-------------+----------------------------
+----------------------------+------+-----+------+------+------+------+---------+-----------+-------------+-----------+----------+-----------+----------------+-----
-------------+-----------+--------+--------+---------+-----+---------------+---------------+-------------------+----------------+-------------+-------------+----
-------------------+
| pid  | name    | path                       | cmdline                                                                           | state       | cwd
| root                       | uid  | gid | euid | egid | suid | sgid | on_disk | wired_size | resident_size | total_size | user_time | system_time | disk_bytes_read | disk
_bytes_written | start_time | parent | pgroup | threads | nice | elevated_token | secure_process | protection_type     | virtual_process | elapsed_time | handle_count | per
cent_processor_time |
+------+---------+----------------------------+-----------------------------------------------------------------------------------+-------------+----------------------------
+----------------------------+------+-----+------+------+------+------+---------+-----------+-------------+-----------+----------+-----------+----------------+-----
-------------+-----------+--------+--------+---------+-----+---------------+---------------+-------------------+----------------+-------------+-------------+----
-------------------+
| 5008 | cmd.exe | C:\Windows\System32\cmd.exe | C:\Windows\system32\cmd.exe /c ""C:\Users\vboxuser\Desktop\fake_malware.bat" " | STILL_ACTIVE | C:\Users\vboxuser\Desktop\
| C:\Users\vboxuser\Desktop\ | 1000 | 544 | -1   | -1   | -1   | -1   | 1       | 5568      | 4755456     | 4825088   | 0        | 15        | 2650           | 0
             | 1772172387 | 3748   | -1     | 3      | 32   | 0             | 0             | PsProtectedTypeNone | 0               | 6           | 71          | 156
250               |
+------+---------+----------------------------+-----------------------------------------------------------------------------------+-------------+----------------------------
+----------------------------+------+-----+------+------+------+------+---------+-----------+-------------+-----------+----------+-----------+----------------+-----
-------------+-----------+--------+--------+---------+-----+---------------+---------------+-------------------+----------------+-------------+-------------+----
-------------------+
osquery>
```

Event Viewer

Event viewer is an in-built software to see the logs of the windows.



In the above image you can see audit failure 4 times which is typing password incorrectly for 4 times. And the event id is 4625. If there are more audit failure we can say that a brute force is happening.

Browser History Analysis

Use Eric Zimmerman's Tools to parse Chrome history for malicious URLs. First install eric zimmerman's tools then install .NET Desktop Runtime 9.x (x64) to run those tools. After that create a temp folder in C drive and copy history file in C:\Users\vboxuser\AppData\Local\Google\Chrome\User Data\Default\History

Now in terminal go to path where you downloaded the tools. In that tools go to SQLECmd and then type the command below,

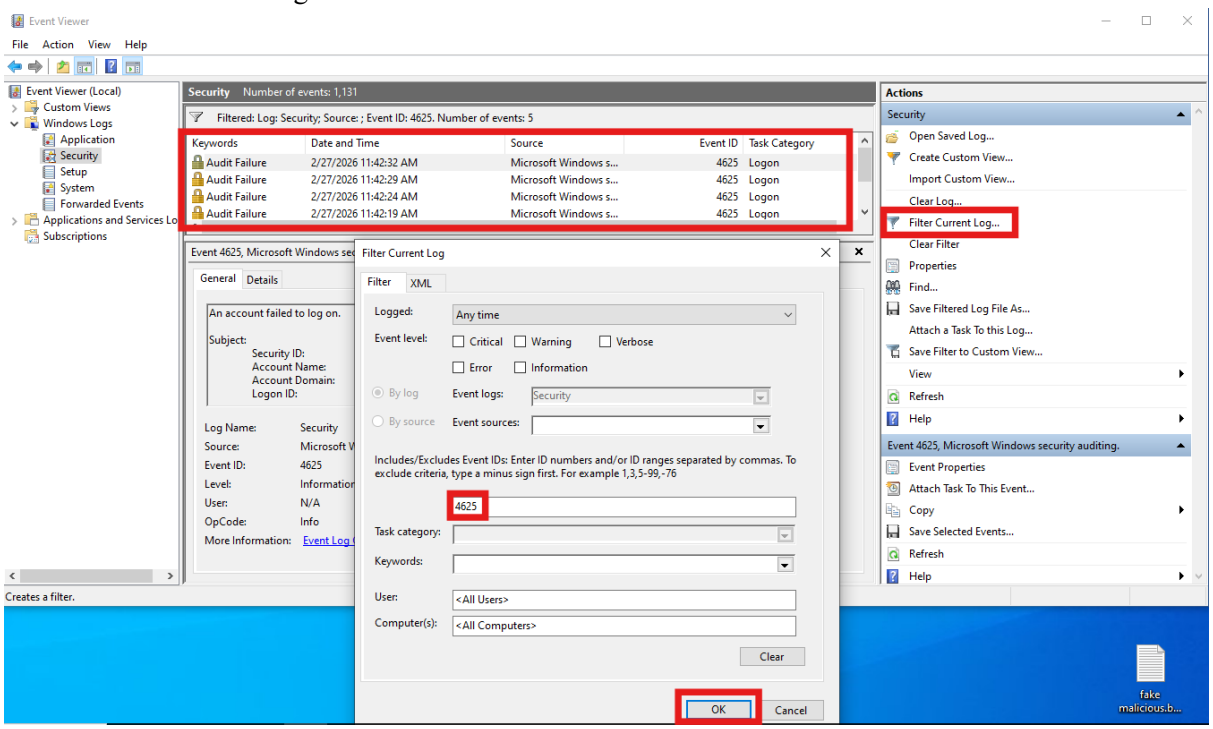
- ./SQLECmd.exe -f C:\temp\History --csv C:\temp\ChromeOutput

```
PS C:\users\tools\net9\SQLECmd> ./SQLECmd.exe -f C:\temp\History --csv C:\temp\ChromeOutput
SQLECmd version 1.1.0.0

Author: Eric Zimmerman (saericzimmerman@gmail.com)
https://github.com/EricZimmerman/SQLECmd
Command line: -f C:\temp\History --csv C:\temp\ChromeOutput

Maps loaded: 92
Path to C:\temp\ChromeOutput doesn't exist. Creating...
Processing C:\temp\History...
        Map queries found: 1. Processing queries...
         Chromium Browser Downloads did not return any results. CSV will not be saved
        Map queries found: 1. Processing queries...
        Dumping Chromium Browser History to C:\temp\ChromeOutput\20260227071924912840_ChromiumBrowser_HistoryVisits_1f37cc07-2b10-4c80-b845-84708e4f7429.csv
        Map queries found: 1. Processing queries...
        Dumping Chromium Browser Keyword Searches to C:\temp\ChromeOutput\20260227071925274956_ChromiumBrowser_KeywordSearches_d968ece4-d24a-4baf-8f07-78da07727712.csv
        For map w/ description Edge Browser History Screenshots, got value 0 from IdentityQuery, but expected 1. Queries will not be processed!


Processed 1 file in 3.9867 seconds

Unable to delete SQLite.Interop.dll. Delete manually if needed

PS C:\users\tools\net9\SQLECmd>
```

You can see that there 2 csv files created, one is for history another is for keyword searches
The output of those 2 files respectively is,
History:

Keyword searches:

Generating failed logins in your Windows VM (e.g., wrong password attempts). Use Event Viewer to filter Event ID 4625 and export results to CSV.
To do this we need first go to lock screen then type some wrong passwords then type correct password and go to event viewer, in that go to windows logs, in that go to security. You can see some failed logins. Now to filter those invalid logins, go to filter current log in right side of the page, then enter event id for failed login then click on ok, now you can see the failed logins in that log. To get more clear understanding there is a screenshot below.



Now to export those logs as CSV, do the following steps, they are,

- With the filtered list displayed, click Action in the top menu bar, then select Save selected events.
- In the "Save as type" dropdown, select CSV (Comma Separated) (*.csv).
- Name the file, choose a location (e.g., Desktop), and click Save.

Create a rule in Wazuh to detect 3+ failed logins in 2 minutes. Test by attempting failed SSH logins (ssh user@192.168.1.x with wrong password).
To do this first you need to go to a path by typing below command,

- sudo nano /var/ossec/etc/rules/local_rules.xml

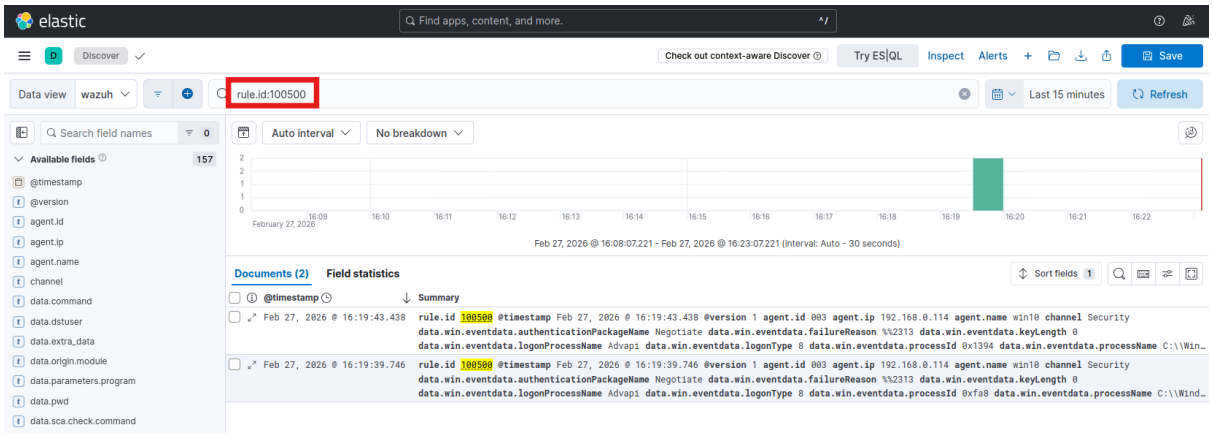
In that xml file, add the below rule,

```
<rule id="100500" level="10" frequency="3" timeframe="120">
  <if_matched_sid>60122</if_matched_sid>
  <description>Windows Brute Force Attack Detected (3 failed logins in 2 minutes)</description>
  <group>windows,authentication,bruteforce,</group>
</rule>
```

After adding that rule, save the file and restart the wazuh service.
Now to trigger this rule, I used ssh from my host machine to windows VM and typed incoorect password.

```
PS C:\Users\saich> ssh vboxuser@192.168.0.114
vboxuser@192.168.0.114's password:
Permission denied, please try again.
vboxuser@192.168.0.114's password:
Permission denied, please try again.
vboxuser@192.168.0.114's password:
vboxuser@192.168.0.114: Permission denied (publickey,password,keyboard-interactive).
PS C:\Users\saich> ssh vboxuser@192.168.0.114
vboxuser@192.168.0.114's password:
Permission denied, please try again.
vboxuser@192.168.0.114's password:
Permission denied, please try again.
vboxuser@192.168.0.114's password:
vboxuser@192.168.0.114: Permission denied (publickey,password,keyboard-interactive).
```

Make sure that ssh service and port is open in the target before trying to login from another machine. And since the windows VM has wazuh agent and it sends it's logs to ubuntu which is wazuh manager, the wazuh manager process those logs with the help of logstash, then it sends those processed logs to elastic, and kibana will show the logs in dashboard.



In the above screenshot as we can see, it is kibana dashboard, which only shows 2 documents because I filtered it for showing the documents which has the rule id of 100500, which is our custom rule. Based on this we can say that we got an alert of detecting multiple logins within a certain timeframe. I didn't want to use wazuh dashboard because I already got kibana and the virtual machine does not have enough specifications to run both complete wazuh stack and ELK stack at the same time.